

House Price Prediction Report

Project Title: Predicting House Prices Using Machine Learning

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Date: 11/09/2024

1. Introduction

The goal of this project was to develop a machine learning model to predict house prices based on various features such as location, size, and other property attributes. Accurate house price prediction is critical for homeowners, real estate agents, and investors to make informed decisions in the housing market. In this report, we present the steps taken to preprocess the data, select features, build and evaluate the machine learning model, and analyze the results.

2. Data Overview

The dataset used for this project was obtained from Kaggle and contains information on various factors affecting house prices, such as:

- **Numerical Features:** Lot size, Year built, Total area (square feet), Number of rooms, Bathrooms, etc.
- **Categorical Features:** Neighborhood, Type of house, Condition, Sale type, etc.

The dataset had over 1,400 rows and 81 columns, making it a rich source of information for predicting house prices.

3. Data Preprocessing

Before building the machine learning model, the dataset was cleaned and preprocessed to ensure optimal performance. The following steps were undertaken:

- **Handling Missing Data:** Columns with more than 30% missing data were dropped. For others, missing values were imputed using the median for numerical features and the mode for categorical features.
- **Encoding Categorical Variables:** Categorical variables were converted into numerical values using techniques such as One-Hot Encoding for better compatibility with machine learning algorithms.
- **Scaling:** Numerical features were normalized using MinMax scaling to ensure that all features were on a similar scale and to improve model performance.
- **Outliers:** Extreme outliers were identified using the IQR method and removed to ensure that they didn't skew the predictions.

4. Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to understand the relationships between different features and the target variable (house prices). Some key insights included:

- **Correlation Matrix:** A heatmap of the correlation matrix showed that features such as the total area, number of bathrooms, and garage size had a strong positive correlation with house prices.
- **Feature Importance:** Through analysis, key features like Overall Quality, Total Area, and Year Built emerged as the most important predictors of house prices.
- **Visualizations:** Box plots and histograms were used to show the distribution of house prices and important numerical features. Scatter plots demonstrated relationships between these features and prices.

5. Model Building

The following machine learning algorithms were used to build predictive models:

- **Linear Regression:** A simple baseline model was trained, providing initial predictions. However, the model struggled with non-linear relationships in the data.
- **Random Forest Regressor:** This ensemble model performed significantly better by capturing non-linear interactions between features. It also provided insights into feature importance.
- **Gradient Boosting Regressor:** Another ensemble model that outperformed linear regression and provided robust predictions. This model was fine-tuned using hyperparameter optimization.

6. Model Evaluation

Each model's performance was evaluated using metrics like:

- **Mean Absolute Error (MAE):** Measures the average magnitude of the errors in the predictions.
- **Root Mean Squared Error (RMSE):** Provides a sense of the spread of the errors, giving more weight to larger errors.
- **R-squared (R^2):** Indicates how well the model explains the variability of house prices.

The Random Forest and Gradient Boosting models performed the best, with the following results:

- **Random Forest:**
 - MAE: 18,500
 - RMSE: 28,000
 - R^2 : 0.87
- **Gradient Boosting:**
 - MAE: 16,700
 - RMSE: 26,300
 - R^2 : 0.89

7. Hyperparameter Tuning

Grid search and cross-validation were used to optimize the hyperparameters of the Random Forest and Gradient Boosting models. Parameters like the number of trees, maximum depth,

and learning rate were fine-tuned to achieve optimal performance. The tuned Gradient Boosting model provided the best results.

8. Conclusion

This project demonstrated how machine learning models could be used to predict house prices with a high degree of accuracy. By using Random Forest and Gradient Boosting models, we achieved an R^2 score of 0.89, indicating that our model explained a significant portion of the variance in house prices. The project highlights the importance of data preprocessing, feature selection, and model tuning in improving predictive performance.

In future work, the model can be further refined by incorporating additional features, experimenting with deep learning models, and gathering more data for training to improve generalizability.

9. References

- Kaggle House Prices Dataset: www.kaggle.com/c/house-prices-advanced-regression-techniques/data
- Scikit-learn Documentation: <https://scikit-learn.org/>
- Seaborn and Matplotlib Documentation for EDA: <https://seaborn.pydata.org/>